

Unit IV - Experiential Learning

1 | Balanced Trees

2 | Expression Trees

2.1 | Given Expression

Consider the mathematical expression in infix notation:

$$(A + B) \times (C - D) / E$$

2.2 | Tasks

1. Convert the expression to postfix notation by following the operator precedence rules.
2. Construct the Expression Tree from the postfix notation.
3. Show the tree after each step of construction, illustrating the process of forming subtrees for operators.
4. Perform and record different tree traversals:

- Preorder Traversal
- Inorder Traversal
- Postorder Traversal

5. Evaluate the Expression Tree for the following values:

$$A = 5, B = 3, C = 8, D = 4, E = 2$$

and compute the final result of the expression.

3 | Red-Black Trees

3.1 | Problem Statement

You are given the following sequence of keys to be inserted into an initially empty Red-Black Tree:

{30, 15, 40, 10, 20, 35, 50, 25, 5, 60}

3.2 | Tasks

1. Construct the Red-Black Tree by inserting the keys one by one while maintaining the Red-Black Tree properties:
 - Every node is either red or black.
 - The root is always black.
 - Red nodes cannot have red children (No two consecutive red nodes).
 - Every path from a node to its descendant NULL nodes must have the same number of black nodes.
 - New nodes are inserted as red, and rotations are used to maintain balance.
2. Show the tree after each insertion, including the necessary rotations and recoloring operations.
3. Determine the height of the final Red-Black Tree.
4. Tabulate the number of rotations performed to maintain balance.

3.3 | Expected Output

- A step-by-step construction of the Red-Black Tree with labeled rotations and recoloring operations.
- The final Red-Black Tree diagram.
- The number of rotations and the final height of the tree.

4 | AVL Trees

4.1 | Given Sequence

Construct the AVL Tree step by step by inserting the following values while performing rotations wherever necessary:

{12, 34, 2, 4, 6, 8, 7, 99, 87, 92, 12, 16, 14, 66, 48, 84, 52, 80}

4.2 | Summary of Rotations

- Number of Right Rotations: ---
- Number of Left Rotations: ---
- Number of Left-Right Rotations: ---
- Number of Right-Left Rotations: ---

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